12C — Deep-Dive Feature Audit: UI, Rendering, Serialization & Master Architecture (Part 3 of 4)

Save/Load Serialization (`SIM.serialize`/`deserialize`), `localStorage` 5 MiB Quota Safety, 40-Script Monkey-Patch Overwrite Bugs, `build_single.js` Offline Gap & `validate_data_schema.js` Blind Spots

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Primary Audited Files**: `js/sim.js`, `test/load.js`, `index.html`, `tools/build_single.js`, `tools/validate_data_schema.js`, `js/ai.js`, `js/cpeco.js`, `js/bottleneck.js`, `js/stats.js`
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)

---

1. `[SYS-ORDER-01]` Critical Monkey-Patch Load-Order Bugs Across the 40 Scripts (`test/load.js` & `index.html`): Later Scripts Silently Erase Earlier Features!

Because `JianghuSect 2` loads `37` logic/data scripts + `3` UI/Render/Main scripts (`40` total) into `globalThis` sequentially, **three later scripts overwrite functions defined in earlier scripts instead of composing them**:

1.1 `cpeco.js:394` (`Script #25`) Completely Overwrites `ai.js:516` (`Script #18`) `AI.nextTechToStudy` — Wiping Out `CULT.techRetentionScore`!

- **Exact Verbatim Proof**:
  - In `js/ai.js:516-544` (`loaded at #18`), `AI.nextTechToStudy` carefully checks `activeKnown.length >= CULT.maxTechs(n)` and requires `CULT.techRetentionScore(n, t) > minReplaceScore + 15` before studying a new active technique when slots are full.
  - Seven scripts later, **`js/cpeco.js:394-413` (`loaded at #25`) redefines `AI.nextTechToStudy = (W, n) => { ... }` from scratch WITHOUT `CULT.techRetentionScore`**!

    - **Chain-Reaction Consequence**: Once a disciple reaches `maxTechs`, they spend `150–600 CP` (`CPE.techPrice`) studying weaker active techniques that `A.study.tick` (`ai.js:499` -> `CULT.pruneExcessActiveTechs`) **immediately deletes upon completion**, burning hundreds of CP in an endless loop!
    - *(Note: `cpeco.js:322` & `331` also completely overwrite dead `A.shop` & `AI.wantedPill` blocks still sitting at `ai.js:602–620`.)*
- **Fix**: Merge `CULT.techRetentionScore` slot-replacement filtering directly into the authoritative `AI.nextTechToStudy` in `cpeco.js:394` (and delete dead shadowed `A.shop`/`AI.wantedPill` in `ai.js:602–620`).

1.2 `stats.js:658` (`Script #24`) Captures `ECO.applyPill` BEFORE `cpeco.js:419` (`Script #25`) Wraps `qiRestore / energyRestore`!

- Because `stats.js` (`#24`) runs `const oApplyPill = ECO.applyPill` **one script before** `cpeco.js` (`#25`) wraps `ECO.applyPill` with `qiRestore`, `energyRestore`, and `restRestore` (`and returns early on` cbBuff `at line 659`), combat pills bypass downstream wrappers (`06A` / `06D`). Move `qiRestore / energyRestore / restRestore` into the base `ECO.applyPill` inside `economy.js` (`#16`)!

1.3 `bottleneck.js:146` (`Script #22`) Overwrites `AI.ACTS.breakthrough.start` From `ai.js:478` (`Script #18`), Erasing `VOW_NO_MEDICINE` Protection!

- `ai.js:481` guards `hasNoMedVow`, **then `bottleneck.js:146` replaces `A.breakthrough.start` completely** and calls `BN.prepare` without checking `CULT.isAbstainingMedicine(n)` (`06C` / `11B`)!

---

2. `[SYS-VAL-02]` Three Massive Blind Spots in `tools/validate_data_schema.js` (`Why 0` Errors`Missed`54`Dead Techniques &`475 `Orphaned Recipes!`)

Inspecting `tools/validate_data_schema.js` (`lines 126–215`) reveals why critical data bugs passed CI silently:

1. **Skipped `data/data_tech.js` (`1,200+` Techniques!) Completely (`lines 169–173`)**:

- Validator checks `JD.MARTIAL.techniques` and `JD.MARTIAL.v200Techniques`, **completely omitting `JD.TECH.techniques` and `JD.TECH.mantras`** (`where the 54 mis-tagged ART190_* *_MANTRA combat techniques live,` 03A`)!

2. **`move.type === 'p'` Typo Exempted `100%` of Physical Attack Moves From `M > 0` Checks (`line 211`)!**:
    - Read line `211` verbatim:

```
JavaScript
const isDefensiveOrBuff = move.arch === 'guard' || ... || move.type === 'p' ||
move.type === 'b' || move.M === 0;
```

In this engine, **`move.type === 'p'` means `Physical Damage` (`vs 'm' = Magical`), NOT Passive**, AND checking `move.M === 0` inside `isDefensiveOrBuff` tautologically exempted broken `M: 0` attacks from failing `move.M > 0`!

3. **Skipped `JD.ECOLOGY.recipes` & Runtime Registry Parity (`lines 126–131`)**:
    - Add `...(JD.ECOLOGY?.recipes || [])` and assert `Object.keys(ECO.recipeById)` === `ECO.recipesFor(W, ['ALCHEMY','FORGING','ARRAY','TALISMAN'], { includeLocked: true })` so no expansion recipe is ever orphaned (`10B`)!

---

3. `[SYS-SAVE-03]` Complete `SIM.serialize` / `SIM.deserialize` Parity & `localStorage 5 MiB` Multi-Slot Quota Audit (`js/sim.js`, `js/main.js`)!

3.1 Master Save/Load Bug Checklist (`Must Be 100% Clean`)

1. **Graveyard String-vs-Object Crash (`sim.js:488` vs `583`, `07E`)**: `W.graveyard` stores string IDs (`'n2_...'`), while `deserialize` checks `g2.id === s.sect.masterId` (`fix: (typeof g2 === 'string' ? g2 : g2?.id) === s.sect.masterId`) and `delete W.byId[n.id]` upon burial!
2. **Non-Enumerable `_vcNorm` Stripped on Reload (`05D`)**: Persist `attunedStars` / `q` scaling without relying on non-enumerable `_vcNorm`, preserving `3★` Relic stats (`1.37×` vs `1.25×` clamp) identically after reload!
3. **Missing Mid-Fight & Persistent State Keys (`sim.js:560`)**:
    - Add `'_elemAuras'`, `'_poiseGauge'`, `'_ccDrStacks'`, `'_shadowClones'`, `'_boneArmorHp'`, and `'_supHealIcd'` to `PERSISTED_UNDERSCORE_KEYS`.

    - Migrate `_btBonusPct`, `_dantianBonusPct`, and `_lastGearSwapAt` to clean persistent fields (`c.btStackBonus`, `c.dantianBonusPct`, `n.lastGearSwapAt`), and exclude RAM-only `W._editUndoStack`!
4. **Missing `.ensure(W)` Hydration Calls on Load**: Invoke `.ensure(W)` in `SIM.deserialize` for `CHARTER_DOCTRINE`, `OUTPOSTS_LOGISTICS`, `STRATEGIC_COUNCIL`, `CARAVAN_AUCTION`, `HEAVENLY_DAO_DIRECTOR`, `MANTRA_SYSTEM`, `DEFENSIVE_ARTS`, `SUPPORT_ARTS`, and `FORBIDDEN_VAULT`.

3.2 Preventing `QuotaExceededError` Across 4 `localStorage` Slots (`jhs_auto` + `jhs_slot_1..3`)

- Browsers enforce a **shared `5 MiB` (`~2.6M` UTF-16 chars) origin quota** across `jhs_auto` + `jhs_slot_1` + `jhs_slot_2` + `jhs_slot_3`. Four 150-year saves with unpruned `relations`, leaked `foeById` summons (`09D`), and verbose `logs` easily exceed `5 MiB`!
- **Fix**: In `SIM.pruneSavePayload`, strip leaked `foeById` summons, prune trivial dead-stranger `relations` (`11D`), exclude `W._editUndoStack` (`12A`), and bound archived `logs` so each serialized save stays compact (`<= 450–650 KB`) even at Year 300!

---

4. `[SYS-BUILD-04]` Offline Standalone Bundler (`tools/build_single.js`) Missing Fallback `wuxia_item_atlas.png`!

- `tools/build_single.js` embeds `world_atlas_terrain.jpg` and the `9` `assets/item_art_[a-z]+.webp` sheets as `data:` URIs, **but forgot to embed `assets/wuxia_item_atlas.png`** (`used by fallback cells` 64–71`and`UI.resolveItemAtlasCell`, verified in` inventory_grid_ui_suite.js:23`)!
- **Exact 3-Line Fix in `tools/build_single.js`**:

```javascript
const legacyAtlas = path.join(root, 'assets', 'wuxia_item_atlas.png');
if (fs.existsSync(legacyAtlas)) {
  html = html.replace(/(['"(])(\.\.\/)?assets\/wuxia_item_atlas\.png(['")])/g,
    (_, q1, _rel, q2) => q1 + '' +
fs.readFileSync(legacyAtlas).toString('base64') + q2);
}
```